

Juneha Hwang

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Education

Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)

Daejeon, South Korea

- **Some relevant courses:** Introduction to Computer Networks, Machine Learning, Graph Machine Learning and Mining, Systems Programming, Computer Organization, Programming Languages

Experience

Visual AI Lab (KAIST)

July 2026–Sep 2026

Internship

Daejeon, South Korea

- Designed and built a multimodal AI agent that answers spatial questions about images and videos through Python execution, integrating vision-language models with object segmentation, tracking, and 3D reconstruction.
- Built reproducible evaluation infrastructure supporting 10+ benchmarks. Used component ablations, human-assisted verification, and execution-trace analysis to identify evidence-selection and error-recovery failures for better agent design.
- Analyzed spatial-reasoning frameworks such as SpatialClaw, SpatialCLI, etc. by inspecting execution traces, segmentation masks, and geometric calculations to trace failures from visual grounding through downstream reasoning.

Linq Alpha

Jan 2025–Feb 2026

Data Engineer

Seoul, South Korea

- Built and maintained private financial-data RAG pipelines for institutional investors, ingesting and structuring earnings calls, broker research, and investment notes for downstream retrieval and LLM workflows.
- Built a parallel-prefix-style algorithm for long-document metadata extraction, propagating preceding context concurrently to reduce latency while preserving context-dependent LLM processing.
- Designed a multi-stage entity-resolution system mapping ambiguous multilingual company references to canonical FIGIs, combining lexical/vector retrieval with ticker, alias, and document-context signals for disambiguation.
- Productionized long-running LLM and data workflows with FastAPI, Celery/Redis, and PostgreSQL, adding asynchronous processing, schema migrations, and observability for reliable operation.

Projects

DEX-DEX Perpetual Futures Arbitrage Bot | *Rust, Tokio, WebSockets*

- Built a low-latency arbitrage bot in Rust that detects and executes spread opportunities between two perpetual futures exchanges across 96 asset pairs in real time.
- Designed a dual-channel architecture using lock-free crossbeam channels for the hot path and Tokio broadcast for cold-path recording, with a dedicated OS thread for the signal engine to avoid async runtime contention.
- Optimized for cache performance with 64-byte aligned price state structs and atomic flags for position tracking, eliminating mutex contention on the critical path.

Papers

FinAgentBench: A Benchmark Dataset for Agentic Retrieval in Financial Question Answering

- C. Choi, J. Kwon, A. Lopez-Lira, C. Kim, M. Kim, **J. Hwang**, et al.
- Published at **ACM ICAIF 2025**, Singapore, Singapore

Structuring the Unstructured: A Multi-Agent System for Extracting and Querying Financial KPIs and Guidance

- C. Choi, A. Lopez-Lira, Y. Lee, J. Kwon, M. Kim, **J. Hwang**, et al.
- Published at **FinIR Workshop @ ACM SIGIR 2025**, Padua, Italy

Honors

Competitive Programming

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025, 2026 Finalist
- International Collegiate Programming Contest (ICPC) Seoul Regional 2025 Finalist, Taichung Regional 2025 Finalist (Silver Medal)

Other Honors

- Jane Street ETC 2024 1st Place
- Fall 2023 Dean's List (KAIST)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, TypeScript, Ruby, SQL, HTML/CSS

Tech Stack: PyTorch, TensorFlow, LangChain, LangGraph, Scikit-Learn, NumPy, Pandas, FastAPI, PostgreSQL, DuckDB, Celery, Redis, Docker, Kubernetes, AWS, React, Git

Languages: Korean (native), English (native), Japanese (conversational)